

- 1) Alter the table STUDENT2 with following structure.

#	Column Name	Constraints
1	RegNo	PRIMARY KEY

```
mysql> USE ddl;
Database changed
mysql> ALTER TABLE student
-> ADD PRIMARY KEY (RegNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> DESC student;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	
Name	varchar(15)	YES	UNI	NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

```
6 rows in set (0.07 sec)
```

- 2) Alter the table name FACULTY with following structure.

#	Column Name	Constraints
1	FacNo	PRIMARY KEY
2	Gender	CHECK 'M' or 'F'

```
mysql> USE college;
Database changed
mysql> ALTER TABLE faculty
-> ADD PRIMARY KEY (FacNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> ALTER TABLE faculty
-> ADD CHECK (Gender IN ('M','F'));
Query OK, 0 rows affected (0.20 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC faculty;
```

Field	Type	Null	Key	Default	Extra
FacNo	varchar(4)	NO	PRI	NULL	
FacName	varchar(15)	YES		NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
MobileNo	bigint	YES		NULL	

```
6 rows in set (0.00 sec)
```

- 3) Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints
1	DeptNo	PRIMARY KEY

```
mysql> ALTER TABLE department
-> ADD PRIMARY KEY (DeptNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> DESC department;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo     | varchar(4)    | NO   | PRI | NULL    |       |
| DeptName   | varchar(15)   | YES  |     | NULL    |       |
| DeptHead   | varchar(4)    | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.05 sec)
```

- 4) Alter the table name COURSE with following structure.

#	Column Name	Constraints
1	CourseNo	PRIMARY KEY
2	SemNo	1 to 6

```
mysql> ALTER TABLE course
-> ADD PRIMARY KEY (CourseNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> ALTER TABLE course
-> ADD CHECK (SemNo BETWEEN 1 AND 6);
Query OK, 0 rows affected (0.13 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC course;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo   | int           | NO   | PRI | NULL    |       |
| CourseDesc | char(15)      | YES  |     | NULL    |       |
| CourseType | char(1)       | YES  |     | NULL    |       |
| SemNo      | int           | YES  |     | NULL    |       |
| HallNo     | int           | YES  |     | NULL    |       |
| FacNo      | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

- 5) Alter the table name STUDENT2 with following structure.

#	Column Name	Constraints
1	S1Name	UNIQUE KEY

```
mysql> USE ddl;
Database changed
mysql> ALTER TABLE student
-> ADD UNIQUE (Name);
Query OK, 0 rows affected, 1 warning (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 1

mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int | NO | PRI | NULL | |
| Name | varchar(15) | YES | UNI | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| City | varchar(15) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

- 6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

```
mysql> ALTER TABLE student
-> MODIFY City INT NOT NULL;
Query OK, 0 rows affected (0.15 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int | NO | PRI | NULL | |
| Name | varchar(15) | YES | UNI | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| City | int | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```